Toward a four-dimensional Yang–Mills construction

Notebook 28: a fixed analytic ball and high-degree stability

Research working document

17 October 2026

Abstract

The generic Taylor-at-zero transport estimate shrinks the background domain at each use. For the composite macrostep of Notebook 27, we avoid this loss by composing inherited activities around a bounded local core and expanding only the distant tail. The pre-gauge representative is small by an inverse macro ratio. A local analytic map supplies strict joint background/fluctuation families for the entire inherited insertion. Cauchy estimates in the fluctuation scaling parameter preserve its small factor in the anchored activity norm. Classical precision and base interactions are substituted into the stationary background before Gaussian integration, using their independent classical domains and finite support reserves. The Gaussian output is then already a function of the next background. A fixed sequence of parameter choices yields one inherited small-field update on the same positive input and output analytic radius, uniformly in preceding refinement and volume. The general activity bound is not contractive. A separate estimate proves an inverse-ratio contraction for background transport when every incoming coefficient has vanishing field Taylor jets through degree four. A bounded Taylor subtraction produces such a remainder. The same estimate, combined with a two-input Gaussian logarithm comparison, contracts the full quantum response to input differences whose coefficientwise low jets agree. Its high-degree output projection can have Lipschitz constant at most one-half. The full update can still regenerate low jets, and this subtraction does not supply a gauge-invariant extraction of them. These results close the stated domain and support interfaces of a finite step and supply a conditional quantum contraction, while leaving uniform quantum amplitudes, coupling flow and the remaining sectors of the existence argument open.

All identities below are first finite-regulator identities on common small domains. Enlarged-domain identities are identified as analytic continuations of those germs. This does not by itself identify a globally continued homogeneous sector with the full Wilson measure or control histories containing large-field regions. The primitive ratio, composite blocking map, gauge choice, retained link/color cutoff and normalization are those constructed in Notebook 27.

1 Inherited interactions on a common background ball

This section proves an anchored estimate for the actual inherited insertion of the composite macrostep. It expands only a supported tail around a bounded local core. The output field domain can be the same product ball as the input domain. This does not assert contraction of activity amplitudes or a complete quantum induction. All scalar identities are on finite compatible periodic regulators; all displayed anchored bounds are uniform in volume and in the preceding classical refinement M.

1.1 An abstract joint-support composition estimate

Let Λ_Q be the old carrier graph and Λ_G the common Gaussian and output-background graph. A projection $\pi : \Lambda_Q \to \Lambda_G$ has fibers of size at most F, takes connected carriers to connected sets, and

has the stated compression bound when used below. Let

$$\sup_i \sum_{H\ni i} e^{a_0|H|}\|E_H\|_b \le \epsilon, \tag{1}$$

where H is connected and E_H is holomorphic in at most $D|H|$ scalar current-field coordinates on the radius-b product polydisc. Write $e_H = \|E_H\|_b$.

The output w lies in a fixed complex product ball, possibly of the same radius b. The retained fluctuation ξ lies in its actual product of real link/color balls. For $|\lambda| < L_*$ suppose the field substituted in E_H has a decomposition

$$f_\lambda|_H = c_{H,\lambda} + v_{H,\lambda}. \tag{2}$$

The core is holomorphic in (w,λ), measurable in ξ, and reads both kinds of variables only in a connected carrier $K_0(H)$ containing πH, where

$$|K_0(H)| \le c_0 + c_1|H|, \qquad K_0(H) \subset N_{h_0}(\pi H), \qquad \|c_{H,\lambda}\|_\infty \le b - m, \quad 0 < m < b. \tag{3}$$

For each old scalar coordinate (i,α) let

$$v_{H,\lambda;i,\alpha} = \sum_X v_{H,\lambda;i,\alpha;X}(w_X, \xi|_{K_0(H)}), \qquad \sup_{H,i,\alpha} \sum_X e^{\kappa\tau(\{\pi i\}\cup X)}\|v_{H,\lambda;i,\alpha;X}\| \le s. \tag{4}$$

The coefficients are strict in their w support X. Their ξ support stays inside the same $K_0(H)$, even when X is remote. The empty X is allowed, and no vanishing at $w = 0$ is required. A stronger complete-fiber norm, with the coordinate maximum inside the support sum, is a sufficient input. All support assignments are fixed independently of λ. Supremum norms include the retained window and smaller closed λ discs. Here τ is graph Steiner length.

Theorem 1.1 (Joint core/tail transport)**.** *Put $t = s/m$ and suppose, for $B \ge 0$ and $\delta > 0$,*

$$t < 1, \qquad B + \delta \le \kappa, \qquad c_1 B + D\log\frac{1}{1-t} \le a_0. \tag{5}$$

There are connected strict joint carriers K with coefficients $U_{K,\lambda}(w_K, \xi_K)$ such that

$$\begin{aligned} \sum_H E_H(f_\lambda|_H) &= \sum_K U_{K,\lambda}, \\ \|U_\lambda\|_B := \sup_z \sum_{K\ni z} e^{B|K|}\|U_{K,\lambda}\| &\le C_T\epsilon, \qquad C_T = e^{Bc_0+\delta h_0} F S_\delta, \\ S_\delta &= \sup_z \sum_y e^{-\delta d(z,y)} < \infty. \end{aligned} \tag{6}$$

The identities and holomorphy follow by normally convergent series on smaller closed parameter domains.

Proof. Expand only in the tail, at the varying local core:

$$E_H(c_{H,\lambda} + v_{H,\lambda}) = \sum_{\nu\in\mathbb{N}^{D|H|}} c_{H,\nu,\lambda}(w,\xi) v_{H,\lambda}^\nu, \qquad \|c_{H,\nu,\lambda}\| \le e_H m^{-|\nu|}.$$

This is scalar-coordinate product Cauchy about a center with the margin in (3). The coefficient reads only $K_0(H)$. No Taylor expansion of the core is made. If needed take Cauchy radii below m and pass to the limit.

Substitute (4) in each scalar occurrence, and join its support X to its root πi by a deterministic minimizing Steiner graph. Let T be the sum of their lengths and K their union with $K_0(H)$. Every ξ dependency already lies in that core carrier. Hence K is a strict connected joint carrier and

$$|K| \le c_0 + c_1|H| + T, \qquad z \in K \implies d(z, \pi H) \le h_0 + T.$$

The anchored weight is bounded by

$$\mathbf{1}_{\{z\in K\}} e^{B|K|} \le e^{Bc_0+\delta h_0} e^{-\delta d(z,\pi H)} e^{Bc_1|H|} \prod_{\text{occurrences}} e^{(B+\delta)\tau}.$$

The absolute sum of support choices and scalar multiindices at fixed H therefore costs exactly

$$e_H e^{Bc_1|H|} \sum_{\nu} t^{|\nu|} = e_H e^{[Bc_1 + D\log(1/(1-t))]|H|}.$$

Repeated support choices already contain their multiplicities; no additional factorial is inserted. Finally $e^{-\delta d(z,\pi H)} \le \sum_{i\in H} e^{-\delta d(z,\pi i)}$. Use (1), the fiber bound and (5). This gives (6), including anchors in the core halo or on added trees. The same positive majorant proves normal convergence. □

1.2 The fluctuation increment retains its small factor

Suppose $L_* \ge 2$ and f_0 is independent of ξ. Use the same canonical support assignment throughout the λ disc and set $Q_K = U_{K,1} - U_{K,0}$. Then

$$\sum_K Q_K = \sum_H [E_H(f_1|_H) - E_H(f_0|_H)], \qquad \|Q\|_B \le \frac{C_T \epsilon}{L_* - 1}. \tag{7}$$

For $1 < \rho < L_*$ take the Cauchy integral in the anchored activity Banach space. The nth coefficient norm is at most $C_T \epsilon \rho^{-n}$. Equivalently, keep the Cauchy integral before the positive anchored sum and bound the whole anchored norm of its integrand. One must not replace this argument by a sum of separate carrier suprema over λ. Let $\rho \uparrow L_*$ and sum $n \ge 1$ at $\lambda = 1$.

The total at $\lambda = 0$ is the inherited background functional, not a vacuum merely because it is independent of fluctuations. In the actual construction below its coefficients can themselves be chosen ξ independent. If $q = C_T\epsilon/(L_* - 1)$, each $\|Q_K\| \le q$ and

$$\sup_z \sum_{K \ni z} e^{B|K|} \|e^{Q_K} - 1\| \le e^q q. \tag{8}$$

The Gaussian multiplier is the bounded extension $\chi_K(\xi)(e^{Q_K} - 1)$, set to zero outside the original retained window on K. On the global window this is exact. Additional indicators are retained separately. For the strict Gaussian theorem it is enough to choose

$$B \ge 4 + \log \mathcal{K}_\theta + \kappa_G, \tag{9}$$

since a connected K has metric spanning-tree length $|K| - 1$. Under the full precision, baseline, tail and inherited budgets, complex interpolation of the inherited exponent gives a connected-log difference bounded by a fixed constant times q. The integrated theorem below supplies this argument with the base exponent retained. This application also requires the precision and all other multipliers to have their strict w supports on this graph.

1.3 The actual local composite fluctuation map

Fix the primitive ℓ. On the final unit graph, an output fiber consists of all current link potentials with initial sites in one R-block, with their maximum norm. Its dimension is of order $dR^d \dim \mathfrak{g}$; it is not given an ℓ^1 coordinate norm. For every prescribed finite reserve A, the supported pre-gauge theorem gives

$$a_{p,i}(w) = \sum_X a_{i;X}(w_X), \qquad \sup_i \sum_X e^{\kappa_0 \tau(\{i\}\cup X)+A|X|}\|a_{i;X}\|_b \le C_A b/R. \tag{10}$$

The complete-fiber maximum is inside the support sum. κ_0, C_A and its domain are fixed at the primitive ratio before choosing $R = \ell^s$, uniformly in s, k. Every coefficient vanishes at $w = 0$.

For an independent small current potential a, define

$$V(a) = e^{ia}, \qquad u(a;y) = 1, \quad u(a;x) = P_R(V(a);y,x), \qquad V^\star(a) = V(a)^{u(a)}.$$

Using the actual composite chart, put

$$\begin{aligned} \Omega_R(a,\zeta)(b) &= \frac{1}{i}\log\Big[e^{\,i\,\mathrm{Ad}_{u(a;b_-)^{-1}}\,\Phi_{V^\star(a)}(\zeta)(b)}\,e^{ia(b)}\Big], \\ \Delta_R(a,\zeta) &= \Omega_R(a,\zeta) - a, \qquad \Delta_R(a,0) = 0. \end{aligned} \tag{11}$$

This is defined for independent a on a fixed R-dependent product ball. Stationarity is only used when $a = a_p(w)$ to identify the physical branch. The map itself is an ordinary local analytic map.

A nonpivot output reads its own retained variable and the local endpoint gauge. A pivot output reads retained variables and backgrounds in its two endpoint R-blocks; all gauges on those links are internal to the same blocks. Thus the whole output fiber at i reads a, ζ in a fixed neighborhood $\mathcal{N}(i)$. The conservative choice $\mathcal{N}(i) = B_2(i)$ contains every needed block. Let $N_0 = \sup_i |\mathcal{N}(i)|$ and $r_0 = 2$, counting periodic multiplicities if necessary.

Finite-dimensional holomorphy, on a slightly larger local domain, and the zero in (11) imply

$$\|\Delta_R(a,\zeta)_i\| \le C_R^{\mathrm{loc}}\|\zeta|_{\mathcal{N}(i)}\|_\infty \quad \text{for} \quad \|a|_{\mathcal{N}(i)}\|_\infty,\ \|\zeta|_{\mathcal{N}(i)}\|_\infty < \rho_R. \tag{12}$$

Integrate the bounded ζ derivative along $t\zeta$ to obtain the linear factor. The complete output maximum is retained. Constants may depend on R, but the map contains no variables at the preceding mesh $1/M$.

1.4 A supported family without summing fine coordinates

Expand only the local Banach background input:

$$\Delta_R(a,\zeta)_i = \sum_{n\ge 0} V_{i,n}(\zeta)[a,\dots,a], \qquad V_{i,n} = D_a^n \Delta_R(0,\zeta)_i/n!.$$

Radial Cauchy bounds the diagonal homogeneous polynomial; complex polarization costs $n^n/n! \le e^n$. After a harmless radius shrink,

$$\|V_{i,n}(\zeta)\|_{\text{multilinear}} \le C_R^{\mathrm{loc}}\|\zeta\|(e/\rho_R)^n. \tag{13}$$

This estimates whole Banach fibers and does not sum their R^d scalar directions. Each coefficient is holomorphic in ζ and zero at $\zeta = 0$.

Decompose each input slot over its at most N_0 neighboring fibers and insert (10). Assign the union of the child supports as the new w support. Intermediate roots are Steiner vertices, so

$$\begin{aligned} \tau(\{i\}\cup\textstyle\bigcup_l X_l) &\le \sum_l [\tau(\{j_l\}\cup X_l) + d(i,j_l)], \\ |\textstyle\bigcup_l X_l| &\le \sum_l |X_l|. \end{aligned} \tag{14}$$

The degree-n absolute rooted norm is therefore bounded by

$$C_R^{\mathrm{loc}}\|\zeta\|\left[\frac{eN_0e^{\kappa_0 r_0}C_Ab}{R\rho_R}\right]^n. \tag{15}$$

Choose b so that the bracket is at most $1/2$. The geometric sum proves a strict family

$$\Delta_R(a_p(w),\zeta)_i=\sum_X d_{i;X}(w_X,\zeta|_{\mathcal{N}(i)}),$$
$$\sup_i\sum_X e^{\kappa_0\tau(\{i\}\cup X)+A|X|}\|d_{i;X}\|\le C_R\|\zeta\|. \tag{16}$$

This keeps the coordinate maximum inside the support sum and bounds each degree before summing; no supremum/order interchange is used. The empty support coefficient $V_{i,0}(\zeta)$ is retained. It is a local fluctuation function independent of w, not a vacuum constant. Every $d_{i;X}$ vanishes at $\zeta=0$.

For the actual retained threshold T set

$$f_\lambda(w,\xi)=\Omega_R(a_p(w),\lambda g\xi). \tag{17}$$

Its background and fluctuation families obey

$$\|a_p\|_{\kappa_0,A;b}\le C_Ab/R,\qquad \|f_\lambda-a_p\|_{\kappa_0,A;b,T}\le C_R|\lambda|\,|g|T. \tag{18}$$

Every output-root coefficient reads ξ only in $\mathcal{N}(i)$, regardless of how far its w support extends. Choose

$$R\ge 8C_A,\qquad L_*=\frac{b}{8C_R|g|T}\ge 2,\qquad L_*|g|T<\rho_R. \tag{19}$$

The last condition is another fixed upper bound on b. On this disc the absolute family norm of f_λ is at most $b/4$. Thus any truncation by its support labels has component amplitude at most $b/4$. At $\lambda=0$ the family is exactly that of a_p, independent of ξ.

1.5 The core halo and the tail after auxiliary grouping

Set $\kappa_t=\kappa_0/2$. Put the labels $X\subset B_{\ell_{\mathrm{tail}}}(i)$ in the core, including the empty support, and all other labels in the tail. Every tail label has $\tau(\{i\}\cup X)>\ell_{\mathrm{tail}}$, hence

$$\sup_i\sum_{\text{tail X}} e^{\kappa_t\tau(\{i\}\cup X)+A|X|}\|f_{\lambda;i;X}\|\le(b/4)e^{-(\kappa_0/2)\ell_{\mathrm{tail}}}. \tag{20}$$

Both families still read ξ only in $\mathcal{N}(i)$. The core's actual component amplitude is at most $b/4$.

Group the source/final-unit cells into centered odd J-blocks, conveniently $J=\ell^j$. These Gaussian macrocells have side JR in current units and become J-blocks of the next current lattice. Old quantum carriers are J-blocks of the current lattice. Their projection π_R to Gaussian macrocells has ratio R and fiber R^d. Both tilings refine the common centered JR tiling.

For any desired fixed macro tail exponent κ,

$$\tau_{\mathrm{macro}}(\{\rho_J i\}\cup\rho_J X)\le 2J^{-1}\tau_{\mathrm{unit}}(\{i\}\cup X)+d|X|.$$

Thus $2\kappa/J\le\kappa_0/2$ and $A\ge d\kappa$ pay the projection. Taking the complete macro output maximum inside the support sum costs at most J^d old output roots, giving

$$s_{\mathrm{tail}}\le J^d(b/4)e^{-(\kappa_0/2)\ell_{\mathrm{tail}}}. \tag{21}$$

Each old J-current input fiber is a restriction of the containing whole JR macro output fiber, so it inherits this bound at its projected macro root. There is no further sum over its scalar coordinates here.

If $\ell_{\mathrm{tail}} + r_0 \le J$, a conservative joint core carrier for a connected old H is

$$K_0(H) = N_{\infty,2}(\pi_R H). \tag{22}$$

It is connected and contains every core w dependency and every ξ dependency of every tail term for outputs in H. In particular

$$|K_0(H)| \le 5^d |\pi_R H|, \qquad K_0(H) \subset N_{2d}(\pi_R H).$$

Declaring the whole halo only enlarges the support. The distant background labels do not bring distant fluctuation dependencies.

Connected-carrier compression at odd $R \ge 5$ gives

$$|\pi_R H| \le 2^d + (2^d/k_R)|H|, \qquad k_R = 1 + \lfloor (R-3)/4 \rfloor.$$

The abstract theorem therefore uses

$$c_0 = 5^d 2^d, \quad c_1 = \frac{5^d 2^d}{k_R}, \quad h_0 = 2d, \quad F = R^d, \quad m = b/2. \tag{23}$$

Its scalar-coordinate count is the actual old quantum-cell value

$$D(J) = dJ^d \dim \mathfrak{g}, \tag{24}$$

not the retained Gaussian dimension.

1.6 A noncircular choice of parameters on the same ball

Fix B using a dimension-independent upper bound in (9), before choosing R, J. For example $\kappa_G = 2$, $m_G = 3$ and the eventual choice $\theta = 1/(4D_G)$ give $\log \mathcal{K}_\theta \le 5/8$ for every D_G, so $B = 53/8$ can be fixed now. The later Gaussian dimension changes θ, not B. Fix $\delta > 0$, choose $\kappa \ge B + \delta$, and choose the finite reserve $A \ge d\kappa$ in (10). Its constants are then fixed at primitive ℓ.

Choose $R = \ell^s$ large enough that

$$R \ge 8C_A, \qquad \frac{5^d 2^d}{k_R} B \le a_0/2. \tag{25}$$

Choose $J = \ell^j$ large enough for the above spatial projection and the separate classical/Gaussian regrouping conditions. With $\ell_{\mathrm{tail}} = c \log J + O(1)$, where $c > 4d/\kappa_0$, one has $\ell_{\mathrm{tail}} + r_0 \le J$ eventually, while

$$D(J)\frac{s_{\mathrm{tail}}}{m} \le \frac{d \dim \mathfrak{g}}{2} J^{2d} e^{-(\kappa_0/2)\ell_{\mathrm{tail}}} \longrightarrow 0. \tag{26}$$

Increase J and the fixed offset so that $s_{\mathrm{tail}}/m \le 1/2$ and $D(J)\log[1/(1 - s_{\mathrm{tail}}/m)] \le a_0/2$. These choices are independent of b: it cancels from the tail-to-margin ratio.

Now choose b once below all classical and chart radii, including (15) and (19). Finally choose $|g|T$ for $L_* \ge 2$ and the desired interaction budgets. The same radius b is the incoming inherited domain and the outgoing background domain. It has not been replaced by a smaller recursive b/C at this step.

The actual map now supplies all the hypotheses of Theorem 1.1. Equations (7) and (19) yield the strict joint bound

$$\|U_1 - U_0\|_B \le \frac{C_T \epsilon}{L_* - 1} \le 16 C_T C_R \frac{|g|T}{b}\epsilon, \qquad C_T = e^{B5^d 2^d + 2d\delta} R^d S_\delta. \tag{27}$$

The exact inherited multiplier norm is at most $e^q q$ with q the first bound in (27). The total U_0 is $E(a_p(w))$, with the same stronger carrier bound $C_T\epsilon$, and can be chosen termwise independent of ξ. It has no g factor and is retained as a background activity, not discarded as a vacuum.

1.7 Scope of the common-domain estimate

The finite-regulator identity and gauge comparison are those of the composite integral module: on the common overlap total gauge invariance identifies f_λ with the actual rooted insertion, while the pre-gauge expression defines its continuation on the common b-ball. No coefficientwise gauge invariance, fine-coordinate ℓ^1 count, or rooted $1/R$ angle contraction is assumed.

The precision and purely classical base multipliers must also be supplied with strict background/joint supports on this graph before applying the Gaussian theorem directly in w. They have independent classical domains and arbitrary finite classical support reserves; their interface is a separate module. Uniform complex nonvanishing follows only after all cluster hypotheses and budgets have been met.

The constants here may be extremely large. The background term has a finite prefactor, not a contraction, and only the fluctuation difference gains $|g|T/b$. Vacuum and marginal extraction, coupling flow, large-field complements, mixed histories and observable bounds remain additional requirements for an existence proof.

2 Classical support substitution before Gaussian integration

This section places the classical precision and base factors directly in the next background w, before applying the Gaussian expansion. The composite ratio $R = \ell^s$ is fixed. Constants may depend on R and on the later auxiliary grouping, but are uniform in the preceding homogeneous depth M and compatible volume. The inputs are the established classical strict families with arbitrary fixed cardinality reserve and the actual supported stationary map. No such arbitrary reserve is assumed for inherited quantum activities.

2.1 A precision composition lemma

On one fixed cell graph let $B(a) = \sum_Y B_Y(a|_Y)$ be a symmetric matrix family. Its matrix rows and columns are supported in Y, and $B_Y(0) = 0$. Each a-cell has D scalar coordinates with independent analytic radius b_{cl}. Write $\tau(Y)$ for graph Steiner-tree length, allowing auxiliary vertices. Suppose

$$\sup_i \sum_{Y \ni i} \sum_{j \in Y} e^{\sigma\tau(Y)+A|Y|} \|(B_Y)_{ij}\|_{b_{\mathrm{cl}}} \le N_B. \tag{28}$$

The norms inside this sum are block operator norms. Let the stationary substitution have a normally convergent strict family

$$a_i(w) = \sum_X u_{i;X}(w_X), \qquad u_{i;X}(0) = 0, \qquad \sup_i \sum_X e^{\kappa\tau(\{i\}\cup X)+a|X|} \|u_{i;X}\|_r \le s(r). \tag{29}$$

Here each output norm is the maximum over the entire cell fiber, *inside* the support sum.

Lemma 2.1 (Composition in the matrix row norm)**.** *Choose $0 < \kappa' < \min(\sigma, \kappa)$, $A' \ge 0$ with $a \ge A'$, and $\rho > 0$. Put*

$$t = \frac{s(r)}{b_{\mathrm{cl}}} < 1, \qquad \alpha_D(t) = D \log \frac{1}{1-t}. \tag{30}$$

If $A' + \alpha_D(t) + \rho \le A$, then $B(a(w)) = \sum_Z B'_Z(w_Z)$ has symmetric strict pieces with rows and columns in Z, and

$$\sup_i \sum_{Z \ni i} \sum_{j \in Z} e^{\kappa'\tau(Z)+A'|Z|} \|(B'_Z)_{ij}\|_r \le N_B \alpha_D(t) \sup_{n \ge 1} n e^{-\rho n}. \tag{31}$$

In particular the bound is $O(r)$ if $s(r) = O(r)$ and the displayed margins are fixed.

Proof. Expand a matrix entry of B_Y on its scalar-coordinate polydisc. Cauchy bounds its multiindex coefficient by $\|(B_Y)_{ij}\|_{b_{\mathrm{cl}}} b_{\mathrm{cl}}^{-|\nu|}$. The zero multiindex is absent. Substitute the strict family (29) in every occurrence of every scalar input. If an occurrence is rooted at $k \in Y$ and assigned support X, declare the resulting carrier to be $Z = Y \cup \bigcup X$. Steiner gluing gives

$$\tau(Z) \le \tau(Y) + \sum_{\text{occurrences}} \tau(\{k\} \cup X), \qquad |Z| \le |Y| + \sum_{\text{occurrences}} |X|. \tag{32}$$

The scalar multiindex sum is

$$(1-t)^{-D|Y|} - 1 \le \alpha_D(t)|Y| e^{\alpha_D(t)|Y|}. \tag{33}$$

The ρ reserve pays $|Y|$ and proves (31). For a fixed nonzero matrix row i, the original piece already had $i \in Y$, and its nonzero columns remain in Y. Substitution does not create new matrix rows or columns. Thus no path-anchor rerooting factor is needed. Symmetry is preserved coefficientwise and after grouping equal Z. Absolute convergence justifies all groupings. □

If only $\sum_Y B_Y(0) = 0$ is initially known, subtract the individual constants. Their absolutely convergent sum is zero, and the input norm increases by at most a factor two. The lemma is a statement about the precision's weighted row norm. It does not assert a norm of whole global matrices anchored at every new support site.

2.2 Application to the actual composite precision

Group current-unit classical families into source R-blocks. The number of current background coordinates in each fiber is finite at fixed R. The established classical support construction, followed by the exact finite-range composite pivot and ordinary logarithm maps, supplies individually symmetric strict pieces of

$$B(a) = A_{M,R}(a) - A_{M,R,0} \tag{34}$$

on an independent current-background product ball. Their row norm has an arbitrary prescribed finite cardinality reserve and some positive spatial exponent, uniformly in M and volume. The chain rule includes the classical current contracted against both chart-curvature terms, and the full absolute gauge Hessian. These terms are not omitted away from the physical branch.

The actual stationary field is the pre-gauge supported representative followed by the fixed-R internal contour gauge. Finite-range Banach-valued analytic composition therefore gives (29) for $a_*(w)$ with any fixed finite cardinality reserve, a positive exponent and $s(r) = O_R(r)$. Its output fiber is the entire current field in an R-block, with its maximum norm inside the support sum. The lemma gives a symmetric strict w-precision with row norm $O_R(r)$. The outer radius b_{cl} here is a classical radius; it is not the inherited quantum radius reused at every step.

Now group source roots and next-background cells into odd J-blocks. For a projected support $\varrho_J Z$, the established geometric bounds give

$$\tau_{\mathrm{mac}}(\varrho_J Z) \le \frac{2}{J}\tau(Z) + d|Z|, \qquad \ell_{\mathrm{mac}}(\varrho_J Z) \le \frac{4}{J}\tau(Z) + 2d|Z|. \tag{35}$$

The second uses metric spanning-tree length ℓ, and is conservative. Grouping matrix rows costs at most J^d: sum the old rows in the macrocell before taking its block norm. For the target strict Gaussian exponent $\gamma = 8$, it is sufficient in $d = 4$ to choose

$$\frac{32}{J} \le \kappa', \qquad A' \ge 64. \tag{36}$$

This produces the required macro precision norm $O_{R,J}(r)$. The separately established covariance decay is regrouped at the same J; increasing J if necessary supplies the fixed Gaussian spatial exponents.

For clarity, the small-radius choice is made *after* R, J. The real baseline retains its spectral bounds $\gamma_R I \le A_{M,R,0} \le B_R I$ under grouping. If $\nu(r)$ bounds the substituted precision perturbation in the macro weighted row norm, then

$$\nu(r) \le \gamma_R/2 \quad \Longrightarrow \quad \|A(w)^{-1} - A_0^{-1}\| \le 2\gamma_R^{-2}\nu(r). \tag{37}$$

Hence the sufficient additional condition

$$\nu(r) \le \frac{\theta\gamma_R^2}{4B_R} \tag{38}$$

puts the covariance within relative norm $\theta/2$ of the real baseline, leaving the other half for the Gaussian Cauchy perturbation. The strict density-support construction has additional positive smallness thresholds in this same weighted precision norm. Since $\nu(r) = O_{R,J}(r)$, all can be met by choosing a single final $b \le r$ after the finite geometry and θ have been fixed. For example $\theta = 1/(4D_G)$ uses the actual finite macro Gaussian dimension D_G. This choice does not require replacing the inherited field radius by a smaller fraction after each stationary substitution.

2.3 Scalar exponents with spectator fluctuation variables

Let $V_Y(a|_Y, \xi|_Y)$ be a classical or measure exponent family, holomorphic in the independent background, measurable on the actual retained cutoff, with

$$\sup_i \sum_{Y \ni i} e^{\sigma\tau(Y)+A|Y|} \|V_Y\|_{b_{\mathrm{cl}},\mathrm{cutoff}} \le \epsilon_{\mathrm{cl}}. \tag{39}$$

Only Y is read in the fluctuation variables. It need not be connected. Empty scalar vacuum terms, if present, are tracked separately. Substitute (29) in the background slots while keeping ξ as spectator data.

Lemma 2.2 (Joint macro carriers for the base factors)**.** *Fix a target cardinality exponent* $\mathsf{B} \ge 0$ *and* $\delta > 0$*. With* $t = s(r)/b_{\mathrm{cl}} < 1$*, suppose*

$$\frac{2(\mathsf{B}+\delta)}{J} \le \min(\sigma, \kappa), \qquad a \ge d(\mathsf{B}+\delta), \qquad A \ge d(\mathsf{B}+\delta) + \alpha_D(t). \tag{40}$$

There is a normally convergent decomposition into connected strict joint (w, ξ) *macro carriers* K *such that*

$$\sum_Y V_Y(a_*(w)|_Y, \xi|_Y) = \sum_K V'_K(w_K, \xi_K), \qquad \sup_z \sum_{K \ni z} e^{\mathsf{B}|K|} \|V'_K\| \le e^{\mathsf{B}} J^d S_\delta \epsilon_{\mathrm{cl}}, \tag{41}$$

where $S_\delta = \sup_z \sum_y e^{-\delta d(z,y)}$.

Proof. Taylor expand in the $D|Y|$ scalar background coordinates only. Retain Y as a fluctuation carrier even in the zero Taylor coefficient. Join $\varrho_J Y$ by a deterministic Steiner graph; for each substituted support X rooted at $k \in Y$, attach a deterministic Steiner graph spanning $\{\varrho_J k\} \cup \varrho_J X$. Their union is a connected joint carrier K. If T is the sum of their edge lengths, then

$$\begin{aligned} &|K| \le 1 + T, \\ &T \le \frac{2}{J}\tau(Y) + d|Y| + \sum_{\text{occurrences}} \left[\frac{2}{J}\tau(\{k\} \cup X) + d|X|\right]. \end{aligned} \tag{42}$$

The carrier includes every background argument and the entire projected fluctuation carrier, regardless of their overlap. For $z \in K$, $d(z, \varrho_J Y) \le T$. Thus

$$e^{\mathsf{B}|K|}\mathbf{1}_{z\in K} \le e^{\mathsf{B}} e^{-\delta d(z,\varrho_J Y)} e^{(\mathsf{B}+\delta)T}. \tag{43}$$

The spatial and cardinal weights in (40) pay the last factor. The scalar multiindex sum, now including degree zero, is exactly $(1-t)^{-D|Y|} = e^{\alpha_D(t)|Y|}$, paid by the remaining input cardinal reserve. Finally use

$$e^{-\delta d(z,\varrho_J Y)} \le \sum_{y\in\varrho_J Y} e^{-\delta d(z,y)}. \tag{44}$$

For each projected root y there are at most J^d source sites. The original anchored norm and the lattice sum S_δ yield (41). These estimates are positive majorants uniform over the actual real cutoff; normal convergence gives background holomorphy. All deterministic support assignments are fixed before choosing the fields. □

The zero background Taylor term can depend on ξ and is then an interaction, not a vacuum constant. The proof does not infer a strict family from a bound on the summed scalar functional. It starts with the actual coefficient bound (39).

2.4 Actual classical and measure inputs and scope

At fixed R, the already established classical support theorem applies to the action after the actual composite chart and ordinary-log conversion. Its Taylor remainder after subtracting constant, linear and quadratic terms has arbitrarily weighted strict coefficients. On the physical stationary branch the subtracted linear term vanishes, so this extension agrees with the exact centered integral. The full absolute gauge action is treated the same way; no off-stationary residual is discarded.

The local ambient pivot logarithms, Haar logarithms and gauge remainder also have actual independent-background coefficient families. Composite one-pivot incidence keeps their ranges finite at fixed R. The retained link cutoff is unchanged. With its link threshold denoted by T, these families retain the respective small factors

$$O_{R,J}(|g|T^3 + |g|T + g^2T^2) \tag{45}$$

after the scalar substitution, provided the fixed chart and Taylor domains hold. Equal-carrier exponents are grouped before exponentiation. If their anchored exponent bound is v, it bounds each individual exponent and gives the multiplier norm bound $e^v v$ for $e^{V'_K} - 1$.

Background-only pivot-delta factors are treated by the same scalar substitution. The Gaussian determinant is expanded in closed chains using the already substituted strict precision; its field-independent normalization remains separate. The classical outer radii and reserves are fixed independently of the old quantum activity radius.

Thus the precision and base multipliers already read w on the auxiliary Gaussian graph. Once a separate inherited joint family meets its budget, the Gaussian expansion outputs w-activities directly on next-current J-cells. There is no second transport through $a_*(w)$ after integration. This changes only the order of normally convergent support substitution; the exact integration variables, covariance, cutoff and normalization are unchanged.

This section is not the integrated quantum-step theorem. It does not estimate inherited activity amplitudes, prove a contractive recursion, identify a running coupling or control large fields. All equality statements start on their common local analytic domains and extend only on the established holomorphic continuations.

3 One composite small-field step on the same analytic ball

We combine Sections 1 and 2 with the actual composite constraint, Gaussian and integral of Notebook 27. All statements are first finite-regulator statements on compatible periodic tori. Constants are uniform in the preceding homogeneous depth $M = \ell^k$ and volume, after the primitive odd ratio ℓ, macro ratio $R = \ell^s$ and auxiliary grouping $J = \ell^j$ are fixed. The theorem concerns a local holomorphic small-field map. The measure identification required beyond its original germ is stated explicitly.

Set $d = 4$ and use the fixed trace-orthonormal real color coordinates. The input activity graph consists of J-blocks of the current unit lattice. The output graph consists of J-blocks of the next lattice, equivalently JR-blocks of the current lattice, and is also the Gaussian integration graph. Thus input and output use the same grouping J in their respective current units. Projection π_R has fiber R^d, preserves connected images, and satisfies

$$|\pi_R H| \le 2^d + \frac{2^d}{k_R}|H|, \qquad k_R = 1 + \left\lfloor \frac{R-3}{4} \right\rfloor . \tag{46}$$

The distinct lifted block interfaces are retained on the chosen cofinal regulator family.

3.1 Statement and exact field-dependent factors

Write the inherited total functional as

$$E(a) = \sum_H E_H(a|_H), \qquad \|E\|_{a_0;b} := \sup_i \sum_{H \ni i} e^{a_0|H|}\, \|E_H\|_b \le \epsilon \le \bar{\epsilon}, \qquad a_0 = \frac{1}{16}, \tag{47}$$

where the carriers are connected and each coefficient is holomorphic on the scalar-coordinate product ball of radius b. The fixed $\bar{\epsilon} < \infty$ bounds the input class. Assume local gauge invariance of the *total* E on its original common domain. If E is a logarithm, its compatible holomorphic branch is part of the input; coefficientwise gauge invariance is not assumed.

Theorem 3.1 (A common-domain composite step)**.** *There are finite choices R, J, positive b, g_0, and retained link/color-ball thresholds $T = T(g) \ge 1$ such that, for $0 < g < g_0$, the chosen small-field macrostep has a holomorphic continuation to the entire next-angle ball $\|w\|_\infty < b$. With $W = e^{iw}$ it has the factorization*

$$\rho_{\mathrm{next}}(W) = N_{M,R,g} \exp\left[-g^{-2} S_{MR}(W) + E_{\mathrm{next}}(w)\right]. \tag{48}$$

Here $N_{M,R,g}$ is field independent, and S_{MR} is the homogeneous ordinary classical minimum at total depth MR, after the $d = 4$ physical rescaling. No coupling renormalization has yet been performed.

The output has connected strict w-activities with the same angle radius b and cardinality exponent a_0. In the notation constructed below,

$$\|E_{\mathrm{next}}\|_{a_0;b} \le C_T \epsilon + (C_\Delta + C_{\det})b + 2S_1 \left[e^3 \tau(T) + e^{v+q}(v+q)\right], \tag{49}$$

$$v \le C_{\mathrm{base}}(gT^3 + gT + g^2T^2), \qquad q \le 16 C_T C_R \frac{gT}{b}\epsilon. \tag{50}$$

All constants are uniform in M and compatible volume. The Gaussian cutoff integral is nonzero on this whole ball and has the analytic logarithm used in (48). The total correction satisfies the local gauge Ward identity throughout the ball, and is invariant along gauge paths that remain in it.

These are statements about the continuation of the actual local small-field germ. They identify the exact chosen macro integral throughout the ball when a gauge-orbit current density with the stated pre-gauge representation is available. They do not identify all quantum histories or the full Wilson pushforward merely from holomorphic uniqueness.

The following construction proves the theorem and specifies its constants and domains. The estimate is not an amplitude contraction: C_T can be much larger than one.

Use the actual composite ordinary map $F_R = \mathcal{M}_\ell^s$ and the independent direct-R contour gauge. The physical backgrounds of Notebook 27 satisfy

$$\begin{aligned} V_p(W) &= \mathcal{M}_\ell^k U_{MR}(W), \qquad V^\star(W) = V_p(W)^{u(W)}, \\ P_R(V^\star) &= 1, \qquad S_M(V^\star) = S_M(V_p) = S_{MR}(W). \end{aligned} \tag{51}$$

The last identity follows from the two actual minimum comparisons. Current spacing is one and the next spacing is R; the coupling-free action has no extra rescaling factor in dimension four.

The retained chart preserves its nonpivot coordinates exactly. The integration field is $e^{i\Phi_{V^\star}(g\xi)}V^\star$ and the window is $\prod_{b \text{ retained}} \mathbf{1}_{\{|\xi(b)|\le T\}}$. The coupling stays inside Φ. Set

$$\begin{aligned} A(w) &= D_z^2 \left[S_M(e^{i\Phi_{V^\star}(z)}V^\star) + \mathcal{G}_R(e^{i\Phi_{V^\star}(z)}V^\star) \right]_{z=0}, \\ E_{\mathrm{pre}}(w) &= E(a_p(w)), \qquad a_p = \frac{1}{i}\log V_p, \\ Z(w) &= \mathbb{E}_{A(w)^{-1}} \left[\chi_{\mathrm{window}}(\xi) e^{V_{\mathrm{base}}(w,\xi)+Q_{\mathrm{inh}}(w,\xi)} \right]. \end{aligned} \tag{52}$$

The base exponent is the centered classical/gauge remainder, the ambient pivot-straightening logarithms, and the Haar logarithms. The inherited difference is exactly

$$\begin{aligned} Q_{\mathrm{inh}}(w,\xi) &= E(f_1(w,\xi)) - E(f_0(w)), \\ f_\lambda(w,\xi)(b) &= \frac{1}{i}\log\left[e^{\,i\,\mathrm{Ad}_{u(w;b_-)^{-1}}\,\Phi_{V^\star(w)}(\lambda g\xi)(b)} V_p(w)(b) \right], \qquad f_0 = a_p. \end{aligned} \tag{53}$$

This group identity changes no integration variable ξ.

For clarity, let n_e count positive current bonds, n_c final macro bonds, n_s noncenter current sites, and $n_z = \dim \mathfrak{g}\,(n_e - n_c)$. In trace-orthonormal exponential coordinates write $d(e^{iX}) = h_0 j(X)dX$, $j(0) = 1$, and put

$$c(g) = \int_G \exp[-(1 - \mathrm{Re}\,\tau(h))/g^2]\, dh.$$

With $\mathcal{D}_R(V)$ the actual pivot-delta determinant and $A_0 = A_{M,R,0}$ the flat retained precision, the field-independent normalization before any additional vacuum subtraction is

$$N_{M,R,g} = c(g)^{-n_s} h_0^{n_e-n_c} g^{n_z} (2\pi)^{n_z/2} \mathcal{D}_R(1) \det(A_0)^{-1/2}, \tag{54}$$

times any separately tracked old field-independent normalization. The background-dependent parts of the pivot and Gaussian determinants remain in E_{next}.

On the common overlap this is precisely the local Haar/delta integral. If an actual gauge-invariant current density on gauge orbits has representation e^{-S_M/g^2+E} at the pre-gauge fields in (53), the same expression is the exact selected sector pointwise throughout the ball. Otherwise it specifies only the verified analytic continuation of that sector's local germ. The measurable completion and one-shot sector are those of Notebook 27, not an asserted intersection of s sequential primitive sectors. Additional original indicators would have to be retained and estimated.

3.2 An explicit order of parameter choices

First fix the exponents

$$a_0 = \frac{1}{16}, \quad \kappa_G = 2, \quad m_G = 3, \quad \gamma = 8, \quad B = \frac{53}{8}, \quad \delta = 1, \quad \kappa_{\mathrm{tail}} = 8. \tag{55}$$

The tuned Gaussian density factor below has logarithm at most 5/8, so (9) holds, and $\kappa_{\mathrm{tail}} \geq B + \delta$.

Choose a fixed pre-gauge cardinal reserve $A_{\mathrm{in}} \geq d\kappa_{\mathrm{tail}}$ before R. By (10), there are $C_A, \kappa_0, r_A > 0$ fixed at the primitive ratio, independently of R, M and volume, with

$$\sup_i \sum_X e^{\kappa_0 \tau(\{i\}\cup X) + A_{\mathrm{in}}|X|} \, \|a_{p,i;X}\|_r \leq C_A r/R, \qquad 0 < r \leq r_A. \tag{56}$$

The norm takes the maximum over the whole current-link fiber of one R-block *inside* the support sum. Choose a sufficiently large power $R = \ell^s$ so that

$$R \geq 8C_A, \qquad \frac{5^d 2^d}{k_R} B \leq a_0/2. \tag{57}$$

The composite chart and its local radii are now fixed. The actual flat retained Gaussian has uniform bounds

$$\gamma_R I \leq A_0 \leq B_R I, \qquad \lambda_0 = B_R^{-1}, \quad \Lambda_0 = \gamma_R^{-1}, \tag{58}$$

and an exponentially local inverse. These constants may depend on R but not on the old depth M.

Next choose any further *classical-only* reserves needed for the fixed-R chart and the two lemmas in Section 2. They may depend on R and use independent classical outer radii. They do not change A_{in} or C_A in (56)–(57). The rooted stationary field is a fixed-R local analytic function of a_p; its support norm is $O_R(r)$ with any such fixed reserve. Lemma 2.1 therefore provides strict symmetric precision pieces with row norm $O_R(r)$.

Choose $J = \ell^j$ sufficiently large for the grouped covariance and precision exponents, in particular

$$\frac{32}{J} \leq \kappa', \qquad A' \geq 64, \qquad \frac{16}{J} \leq \kappa_0/2, \tag{59}$$

with the available κ', A' in (36). Set $\ell_{\mathrm{tail}} = c \log J$ plus a fixed offset, with $c > 4d/\kappa_0$, so that eventually $\ell_{\mathrm{tail}} + 2 \leq J$. The actual joint family of Section 1.5 then has

$$\begin{aligned} s_{\mathrm{tail}} &\leq J^d (b/4) e^{-\kappa_0 \ell_{\mathrm{tail}}/2}, & m_{\mathrm{core}} &= b/2, \\ c_0 &= 5^d 2^d, & c_1 &= 5^d 2^d / k_R, \\ h_0^{\mathrm{halo}} &= 2d, & F &= R^d, \qquad D(J) = d \, \dim \mathfrak{g} \, J^d. \end{aligned} \tag{60}$$

Here h_0^{halo} is a graph radius, distinct from the Haar constant in (54). Increase J and the tail offset until, with $t = s_{\mathrm{tail}}/m_{\mathrm{core}}$,

$$t \leq \tfrac{1}{2}, \qquad D(J) \log \frac{1}{1-t} \leq a_0/2. \tag{61}$$

This is possible because its sufficient bound is a constant times $J^{2d} e^{-\kappa_0 \ell_{\mathrm{tail}}/2} \to 0$. The ratio t in this sufficient estimate is independent of b. The entire local ξ collar and the core lie in the fixed two-neighbor halo used in (22).

Only now choose the final field radius. The macro retained dimension and tunable density parameters are

$$\begin{aligned} D_G &= J^d d (R^d - 1) \dim \mathfrak{g}, & \theta &= \frac{1}{4 D_G}, \\ \mathcal{K}_\theta &= \left[\frac{1+\theta}{(1-\theta)^3} \right]^{D_G/2}, & \log \mathcal{K}_\theta &\leq \frac{5}{8}, \qquad \rho_\theta = \frac{1+\theta}{1-\theta}. \end{aligned} \tag{62}$$

The uniform Gaussian bounds fix a positive common strictly local threshold η_*. Shrink it if necessary so that $\eta_* \leq 1$.

Choose $b > 0$ below all classical, physical-branch and chart radii. In particular require

$$\frac{eN_0 e^{2\kappa_0} C_A b}{R\rho_R} \le \frac{1}{2}, \tag{63}$$

as in (15). The substituted macro precision norm $\nu(b) = O_{R,J}(b)$ must meet every strictly local support-algebra threshold. Include

$$\nu(b) \le \gamma_R/2, \qquad \nu(b) \le \frac{\theta\gamma_R^2}{4B_R}, \tag{64}$$

as justified in (37)–(38). Also impose the tunable density-algebra conditions $u_G \le \theta$ and $w_G \le \theta/[(1+\theta)\Lambda_0]$, where u_G, w_G are its continuous $O(\nu)$ support-algebra quantities, and the smallness for the determinant chain expansion below. All hold after one shrink of b with R, J fixed. This selects a single input and output radius, rather than dividing the inherited radius during substitution.

A bad macrocell for the actual link/color-ball window implies that its full Euclidean retained norm exceeds T. Its valid joint-tail constant is therefore

$$\tau(T) = \mathcal{K}_\theta \, 2^{D_G/2} \exp\left[-\frac{T^2}{4\rho_\theta \Lambda_0}\right]. \tag{65}$$

Choose $T \ge 1$ with $e^3\tau(T) \le \eta_*/4$. This estimates the bad event without replacing the actual good window by a Euclidean macro ball.

Finally require the coupling to be small enough that

$$L_* = \frac{b}{8C_R g T} \ge 2, \qquad L_* g T < \rho_R, \tag{66}$$

and the interaction bounds below hold uniformly for $\epsilon \le \bar{\epsilon}$. The second condition is an additional fixed restriction on b. For instance $T(g)$ may exceed the fixed threshold and grow as a fixed multiple of $\sqrt{\log(1/g)}$. Then $\tau(T) \to 0$, $gT^3 \to 0$ and $gT/b \to 0$ for this fixed b.

3.3 Strict coefficients are already in the next background

Lemma 2.1 composes the actual precision pieces with the supported stationary field while keeping their matrix rows and columns in the original carrier. Its scalar-coordinate factor is paid by the independent classical cardinality reserve. Lemma 2.2 keeps the whole spectator ξ carrier even at zero background Taylor degree, joins each substituted background support to it, and reroots the resulting connected carrier. The classical action includes its pair-supported quadratic part; its actual chart Hessian retains both constraint and ordinary-log curvature terms and the full absolute gauge Hessian.

Thus (45) supplies, directly in (w, ξ),

$$\|V_{\text{base}}\|_B \le v, \qquad v = C_{\text{base}}(gT^3 + gT + g^2T^2). \tag{67}$$

The independent-background definition subtracts constant, linear and quadratic Taylor terms before composition. At the physical branch the total linear term is zero; it is not silently dropped away from that branch.

For the inherited term, the actual complete-fiber construction (16)–(18) yields

$$\|f_\lambda - a_p\|_{\kappa_0, A_{\text{in}}; b, T} \le C_R|\lambda| g T. \tag{68}$$

Every coefficient reads ξ only in the fixed output-root collar, independent of its remote w support. On (66), the whole absolute family has amplitude at most $b/4$. Hence every truncated core has that same bound without using cancellation. Theorem 1.1 and (27) give canonical connected carriers

$$\sum_K U_{K,\lambda} = E(f_\lambda), \qquad \|U_\lambda\|_B \le C_T\epsilon,$$
$$C_T = e^{B5^d2^d+2d}R^dS_1, \qquad S_1 = \sum_{z\in\mathbb{Z}^d} e^{-|z|_1},$$
$$\|Q_{\mathrm{inh}}\|_B \le q := \frac{C_T\epsilon}{L_*-1} \le 16C_TC_R\frac{gT}{b}\epsilon. \tag{69}$$

The $\lambda = 0$ family is termwise ξ independent and represents E_{pre}. Empty background support terms of the fluctuation difference remain interactions if they depend on ξ. Banach Cauchy is applied with its contour integral before the positive anchored sum, as in (7).

All these coefficients are already functions of the next background w on the Gaussian graph. The Gaussian output will need no second stationary substitution and no second generic inherited-radius estimate.

3.4 Nonvanishing and a bound for the inherited logarithm difference

Combine equal-carrier exponents. Their exact multiplier norm is

$$\sup_z \sum_{K\ni z} e^{B|K|}\left\|e^{V_{\mathrm{base},K}+Q_K}-1\right\| \le e^{v+q}(v+q). \tag{70}$$

Products on distinct overlapping carriers are handled by the overlap theorem. Each multiplier is defined on its actual carrier window and extended by zero outside it. This is exact on the global retained window. Since $B \ge 4+\log\mathcal{K}_\theta+\kappa_G$ and connected K has metric spanning-tree length $|K|-1$, this norm pays the strict Gaussian multiplier weight.

Impose

$$v \le \frac{\eta_*}{16e}, \qquad q \le \frac{\eta_*}{32e}, \qquad e^3\tau(T) \le \eta_*/4. \tag{71}$$

With $\eta_* \le 1$, (70) is below the interaction quarter budget. The precision conditions have already been imposed. The tunable strictly local Gaussian theorem therefore proves nonvanishing of Z on the full b-ball and an analytic connected logarithm with

$$a_G = \min\{1/2, \kappa_G/4\} = \frac{1}{2},$$
$$\|\log Z\|_{a_G;b} \le 2S_1\left[e^3\tau(T)+e^{v+q}(v+q)\right]. \tag{72}$$

This is a nonvanishing theorem on a volume-independent complex domain, not an inference from real positivity.

There is also a precise inherited-log difference estimate. For $q>0$, replace Q_{inh} by ζQ_{inh} and set

$$L_Q = \frac{\eta_*}{16eq} \ge 2. \tag{73}$$

On $|\zeta| < L_Q$, the total exponent norm is at most $\eta_*/(8e)$; since $\eta_* \le 1$, its multiplier norm is at most $\eta_*/8$. Together with the bad-block budget, the Gaussian bound is at most $S_1\eta_*$ on that disk. The canonical connected-log coefficients are holomorphic as an anchored Banach family. Cauchy therefore

gives

$$\|\log Z - \log Z_{\mathrm{base}}\|_{a_G;b} \le \frac{S_1\eta_*}{L_Q - 1} \le 32eS_1 q \le 512eS_1C_TC_R\frac{gT}{b}\epsilon. \tag{74}$$

For $q = 0$ the difference is zero. This interpolation keeps the base exponent throughout; it does not omit a factor $e^{V_{\mathrm{base},K}}$ when comparing equal-carrier multipliers.

3.5 External normalizations and the exact update

The pivot and Gaussian determinant corrections are

$$\Delta(w) = \log\frac{\mathcal{D}_R(V^\star(w))}{\mathcal{D}_R(1)}, \qquad Q_{\det}(w) = -\frac{1}{2}\log\det[A(w)A_0^{-1}]. \tag{75}$$

The first is the sum of actual local pivot log determinants composed with the supported stationary field. The scalar composition lemma gives connected coefficients with any required fixed reserve and norm at most $C_\Delta b$.

For the second, put $C_0 = A_0^{-1}$ and $\mathcal{B}(w) = A(w) - A_0$ and expand afresh on the macro graph:

$$Q_{\det}(w) = \frac{1}{2}\sum_{n\ge1}\frac{(-1)^n}{n}\operatorname{Tr}[(C_0\mathcal{B}(w))^n]. \tag{76}$$

Join every entire precision label and each covariance path by deterministic trees and shortest paths. Their cyclic union is connected and has cardinality at most the total edge-length budget plus one. The weighted row norm at $\gamma = 8$ and a rerooting spare give

$$\|Q_{\det}\|_{1/2;b} \le \frac{D_G e^{1/2}S_1}{2}\frac{K_8\nu_8}{1 - K_8\nu_8} \le C_{\det}b, \qquad K_8\nu_8 < 1. \tag{77}$$

Here K_8 bounds the macro baseline covariance and ν_8 the strict precision perturbation at that weight, uniformly in M and volume. Their smallness was included in the choice of b.

With the field-independent factor (54), the exact correction is

$$E_{\mathrm{next}} = E_{\mathrm{pre}} + \Delta + Q_{\det} + \log Z. \tag{78}$$

Its first term has weight B, and the other terms have weight at least $1/2$, all on the same product ball b. They therefore have the required weight $a_0 = 1/16$. Equations (69), (72) and (77) prove (49). Subtracting each output coefficient's value at $w = 0$ costs at most a factor two in the norm and changes only the field-independent normalization.

3.6 Gauge covariance of the sector and Ward continuation

Take a small real coarse gauge k and lift it to current sites as a constant on each final R-block. A background with $P_R = 1$ keeps that property under this lift. The composite constraint transforms from W to W^k, and the classical action is invariant on the common comparison domain. Uniqueness of the nearby physical constrained minimum thus gives

$$V^\star(W^k) = V^\star(W)^{\text{ block-constant lift of } k}. \tag{79}$$

The relative retained chart transforms by source-endpoint adjoints: this follows from constraint covariance, unchanged pivots and uniqueness for the prescribed retained values.

For real compact gauges these adjoints are orthogonal in the chosen color norm. They preserve each actual link/color-ball cutoff and have unit absolute retained Jacobian. The original Haar/delta formula, equivalently its exact pulled-back factors, therefore gives equality of the chosen small-field integrals at W and W^k on a neighborhood of the identity. The inherited identity is an identity of the total functional with its compatible branch. No coefficientwise gauge invariance is used.

The extracted classical action in (51) is also gauge invariant there. Nonvanishing gives the analytic logarithm on the whole product ball. Differentiating the real near-origin identity gives the infinitesimal coarse Ward identity for E_{next}. Its coefficients are holomorphic functions of w. Uniqueness on the real slice, followed by the holomorphic identity theorem on the connected finite-volume ball, extends this identity throughout $\|w\|_\infty < b$. Integrating it gives invariance along gauge paths staying in this ball.

This does not compare arbitrary endpoints whose connecting gauge path leaves the domain. It also does not turn the continued germ into a globally identified current density without the orbit representation specified above. A field-independent vacuum normalization leaves the Ward identity unchanged.

3.7 Finite-step scope

The common radius is obtained by placing the inherited argument at the actual pre-gauge representative and expanding only its distant supported tail. All new classical and measure coefficients are already in w before integration. The Gaussian output therefore incurs no second stationary transport. This proves the stated domain, strict-support and nonvanishing interfaces of one chosen composite step, uniformly in preceding homogeneous refinement and volume.

The estimate still contains $C_T\epsilon$, not a contractive amplitude update. Apart from optional vacuum normalization, no relevant or marginal extraction has been performed. A repeated quantum construction requires a stable extraction and coupling scheme, amplitude control, large-field and mixed-history estimates, compatible global measure identities, and the observable and continuum limits. None follows here from analyticity or the local Ward identity alone.

4 A contraction for the coefficientwise high-field-degree background

The following result concerns only the inherited background $E(a_p(w))$. It uses the actual pre-gauge representative and the same incoming and outgoing activity exponent and field radius. It does not identify a gauge-invariant marginal extraction.

Fix $a_0 > 0$ and a primitive odd ℓ. Old carriers H are connected sets of current J-cells; output carriers are J-cells of the next current lattice, or current JR-cells, where $R = \ell^s$ and $J = \ell^j$. The projection has fiber R^d and satisfies

$$|\pi_R H| \le 2^d + \frac{2^d}{k_R}|H|, \qquad k_R = 1 + \left\lfloor \frac{R-3}{4} \right\rfloor. \tag{80}$$

Use compatible finite regulators. Write

$$\|E\|_{a_0,b} = \sup_i \sum_{H \ni i} e^{a_0|H|}\|E_H\|_b, \tag{81}$$

where each coefficient is holomorphic on its scalar-coordinate radius-b product polydisc. Each old cell has $D(J) = dJ^d \dim \mathfrak{g}$ scalar coordinates.

4.1 Uniform transport on a larger field-scaling disk

Choose $\kappa \geq a_0 + 1$ and a finite classical reserve $A \geq d\kappa$ before choosing R. The established pre-gauge theorem gives constants $C_A, \kappa_0, r_A > 0$, fixed at primitive ℓ independently of R, M and volume, with

$$a_{p,i}(w) = \sum_X a_{i;X}(w_X), \qquad a_{i;X}(0) = 0, \qquad \sup_i \sum_X e^{\kappa_0 \tau(\{i\}\cup X) + A|X|} \|a_{i;X}\|_r \leq C_A r/R \quad (0 < r \leq r_A). \tag{82}$$

The complete current-field maximum in one source R-block is inside the support sum. Put

$$L_* = \frac{R}{8C_A} \geq 2, \qquad 0 < b \leq \frac{r_A}{L_*}, \qquad f_\lambda(w) = a_p(\lambda w), \quad |\lambda| < L_*. \tag{83}$$

The absolute rooted norm of f_λ on the b-ball is at most $b/8$. Every support truncation therefore has that same component bound without cancellation.

Cut the rooted supports at distance ℓ_{tail}, using exponent $\kappa_0/2$ for the tail, and group roots by J. Provided $2\kappa/J \leq \kappa_0/2$, its macro norm obeys

$$s_{\text{tail}} \leq J^d (b/8) e^{-\kappa_0 \ell_{\text{tail}}/2}. \tag{84}$$

The factor J^d is the sum over old output roots before the macro-fiber maximum. Restriction to an old J-current input fiber introduces no additional coordinate sum. For $\ell_{\text{tail}} \leq J$ use the conservative connected core $K_0(H) = N_{\infty,2}(\pi_R H)$. Its constants are

$$c_0 = 5^d 2^d, \qquad c_1 = \frac{5^d 2^d}{k_R}, \qquad h_0 = 2d, \qquad F = R^d, \qquad m = b/2. \tag{85}$$

Choose R so that $c_1 a_0 \leq a_0/2$. Choose J large and $\ell_{\text{tail}} = \lceil c \log J + C \rceil$ with $c > 4d/\kappa_0$, so that $\ell_{\text{tail}} \leq J$ and

$$t := s_{\text{tail}}/m \leq \tfrac{1}{2}, \qquad D(J) \log \frac{1}{1-t} \leq a_0/2. \tag{86}$$

This is possible because $D(J)t \leq (d \dim \mathfrak{g}/4) J^{2d} e^{-\kappa_0 \ell_{\text{tail}}/2} \to 0$. Larger J required by other fixed-step modules can be used. These conditions do not depend on b.

The joint core/tail theorem, specialized to no fluctuation variables and target exponent a_0, now gives a canonical connected family U_λ with

$$\sum_K U_{K,\lambda}(w_K) = \sum_H E_H(a_p(\lambda w)|_H), \qquad \|U_\lambda\|_{a_0,b} \leq C_0 R^d \|E\|_{a_0,b}, \quad C_0 = e^{a_0 5^d 2^d + 2d} S_1. \tag{87}$$

Here $S_1 = \sum_{z \in \mathbb{Z}^d} e^{-|z|_1}$ also bounds the periodic sum. Support choices are fixed independently of λ. The family is holomorphic in the anchored activity Banach space. The prefactor C_0 is independent of J, R, M and volume; R^d is explicit. Although the background substitution is nonlinear, this canonical construction is linear in E.

4.2 Coefficientwise fifth-order vanishing and contraction

Theorem 4.1 (High-field-degree background transport)**.** *Assume, for every input carrier H, the full total field Taylor tensors satisfy*

$$D^n E_H(0) = 0, \qquad n = 0, 1, 2, 3, 4. \tag{88}$$

This includes mixed derivatives in all coordinates of H. Under the preceding parameter choices, the connected output $\mathcal{T}_{R,J} E := U_1$ has the same coefficientwise vanishing jets and obeys

$$\|\mathcal{T}_{R,J} E\|_{a_0,b} \leq C_0 (8C_A)^5 R^{d-5} \|E\|_{a_0,b}. \tag{89}$$

In $d = 4$, for any prescribed $0 < q_{\mathrm{contr}} < 1$, taking the finite power $R = \ell^s$ sufficiently large that

$$R \geq C_0(8C_A)^5/q_{\mathrm{contr}} \tag{90}$$

in addition to the earlier conditions makes this a genuine q_{contr} contraction. It is uniform in old depth and compatible volume.

Proof. In the canonical expansion, a term of tail Taylor degree $q = |\nu|$ contains $(\partial^\nu E_H)(c_\lambda)/\nu!$ and q supported tail factors. Core and tail both vanish at $\lambda = 0$ termwise. For $q \leq 4$, condition (88) makes the derivative at the core vanish to order at least $5 - q$ in λ, and the tail factors supply order q. For $q \geq 5$ the tail factors alone supply order at least five. This includes the core term $q = 0$. Normal convergence on smaller disks permits differentiation and grouping: every canonical carrier, and hence U_λ as a Banach-valued function, has derivatives of orders zero through four equal to zero. No cancellation between carriers is used. The same argument in the field variable w proves the claimed output jet condition.

For completeness, if a Banach-valued holomorphic F on $|\lambda| < L$ is bounded by M and has a zero of order p at zero, then

$$\|F(\lambda)\| \leq M(|\lambda|/L)^p. \tag{91}$$

The singularity of $F(\lambda)/\lambda^p$ is removable. On a circle of radius $\rho < L$ its norm is bounded by M/ρ^p; apply the scalar maximum principle to every bounded linear functional and send $\rho \uparrow L$. Applying this with $p = 5$ to (87), at $\lambda = 1$, gives $C_0 R^d L_*^{-5}\|E\|$, which is (89). A coefficientwise Cauchy remainder bound would contain an unnecessary $(1 - L_*^{-1})^{-1}$ factor. □

The parameter order is noncircular: choose the finite reserve and its constants, then R, then J and its tail truncation, and finally the single radius $b \leq 8C_A r_A/R$ together with any other fixed classical restrictions. The same b is used at the input and output. For the conservative halo in $d = 4$ and $a_0 = 1/16$, $C_0 = e^{633}S_1$; no practically sized scale is asserted. This result concerns the background transport, not the new quantum correction or the inherited fluctuation increment.

4.3 A bounded Taylor subtraction, with its limitations

On the circular product polydisc, define the total homogeneous Taylor projection by

$$(P_n E_H)(a) = \frac{1}{2\pi}\int_0^{2\pi} e^{-int} E_H(e^{it}a)\, dt. \tag{92}$$

It is the total-degree-n Taylor polynomial, has carrier H, and satisfies $\|P_n E_H\|_b \leq \|E_H\|_b$. The phase integral estimates the complete homogeneous polynomial without summing its coordinate monomials. Consequently

$$\mathcal{R}_{\geq 5} E_H = E_H - \sum_{n=0}^{4} P_n E_H, \qquad \|\mathcal{R}_{\geq 5} E\|_{a_0,b} \leq 6\|E\|_{a_0,b}. \tag{93}$$

No dimension factor, radius shrink or cardinal reserve is needed for this projection. The remainder meets (88), and in $d = 4$

$$\|\mathcal{T}_{R,J}\mathcal{R}_{\geq 5} E\|_{a_0,b} \leq \frac{6C_0(8C_A)^5}{R}\|E\|_{a_0,b}. \tag{94}$$

This is not a gauge-invariant extraction scheme. Local gauge transformations act nonlinearly and affinely in a link-potential chart, so the low-degree part and high-degree remainder of a gauge-invariant total need not be separately gauge invariant. The rooted/pre-gauge identity applies to the complete inherited functional before such a split; it cannot be applied separately to these pieces without another argument.

Nor are the low jets automatically a finite-dimensional space of couplings. They include kernels on arbitrarily many carrier geometries and separations. Classifying their local gauge-invariant content, separating low momentum from field degree, constructing nonlinear Ward-compatible completions, and bounding marginal or relevant terms remain separate requirements. New quantum coefficients may regenerate low jets. The contraction proved here therefore supplies an irrelevant-background estimate, not a quantum induction, running coupling or continuum construction.

5 High-jet differences of the actual quantum step

The background contraction can be combined with the actual common-domain fluctuation integral. The result compares two admissible complete inherited inputs. It does not assert that the quantum step preserves their common low jets.

Use the graphs, radius b, exponent $a_0 = 1/16$ and parameters of Theorem 3.1, with the additional choices below. Pad absent coefficients by zero so that both inputs have one common connected-carrier index set. Write

$$E_j = \sum_H E_{j,H}, \qquad \|E_j\|_{a_0;b} \le \bar{\epsilon}, \qquad D = E_1 - E_2, \qquad D_H = E_{1,H} - E_{2,H}. \tag{95}$$

Both total inputs satisfy the local gauge and logarithm-branch hypotheses of (47). Assume the full coefficientwise condition

$$D^n D_H(0) = 0 \qquad (H \text{ arbitrary},\ 0 \le n \le 4), \tag{96}$$

including every mixed field derivative. Equality of low jets only after summing carriers is not sufficient.

5.1 Compatible coefficient families

Choose one pre-gauge reserve large enough for the stronger joint exponent $B = 53/8$, and use its same supported $a_p = \sum_X a_{p,i;X}$ in Sections 1 and 4. Fix one sufficiently large J, one ℓ_{tail}, the same support truncation $X \subset B_{\ell_{\text{tail}}}(i)$, the same halo $K_0(H) = N_{\infty,2}(\pi_R H)$, and the same deterministic joining trees and carrier grouping. The stronger joint budgets imply the weaker background budgets. Also impose the final-radius condition

$$L_H = \frac{R}{8C_A} \ge 2, \qquad bL_H \le r_A. \tag{97}$$

In the actual fluctuation family each coefficient has the form $a_{p,i;X} + d_{\lambda;i;X}$ with $d_{0;i;X} = 0$ termwise. Consequently the core/tail expansion at fluctuation parameter zero is exactly the background expansion at field-scaling parameter one:

$$U^{\text{fluct}}_{K,0}[E] = U^{\text{field}}_{K,1}[E] = (\mathcal{T}_{R,J}E)_K. \tag{98}$$

The local coefficient functions, multiindices, support occurrences and trees coincide before summing K. The two proofs may estimate this same assignment in different weights. Extra sites of the halo can be retained as declared support even when the background coefficient does not read them.

Core/tail differentiation of E_H and support grouping are linear in the input family. Thus $\mathcal{T}_{R,J}$ and the inherited joint difference Q_E are linear, and

$$Q_{E_1} - Q_{E_2} = Q_D, \qquad \|Q_E\|_B \le a_Q \|E\|_{a_0;b}, \qquad a_Q = 16C_T C_R \frac{gT}{b}. \tag{99}$$

This is the actual estimate (69).

For the Gaussian logarithm retain all possible carrier labels, including coefficients that happen to vanish. Use common forest and hull assignments, precision pieces and the normally convergent cluster-log branch. Let $\mathcal{L}(Q)$ denote this one analytic coefficient-valued logarithm map with the fixed base exponent and precision. The compatible actual output family is

$$\mathscr{F}(E) = \mathcal{T}_{R,J}E + \Delta + Q_{\det} + \mathcal{L}(Q_E). \tag{100}$$

Its scalar sum is precisely (78), with the same germ or gauge-orbit density interpretation. The classical background, covariance, cutoff, base factors and external determinant families are identical for the two inputs. No E-dependent quantum saddle is used.

5.2 A two-input Lipschitz estimate

Set

$$C_H = C_0(8C_A)^5, \qquad C_{\text{quant}} = 1024eS_1C_TC_R, \tag{101}$$

where C_0 is (87). It is fixed before R; C_{quant} is fixed after the finite geometry is selected. Strengthen the allowed budgets to

$$v \le \frac{\eta_*}{64e}, \qquad a_Q\bar{\epsilon} \le \frac{\eta_*}{128e}, \qquad e^3\tau(T) \le \eta_*/4, \qquad \eta_* \le 1. \tag{102}$$

All precision and domain conditions of Theorem 3.1 remain in force.

Theorem 5.1 (Quantum differences with common low jets)**.** *For the compatible families above, condition* (96) *implies*

$$\|\mathscr{F}(E_1) - \mathscr{F}(E_2)\|_{a_0;b} \le \left(\frac{C_H}{R} + C_{\text{quant}}\frac{gT}{b}\right)\|E_1 - E_2\|_{a_0;b}\,. \tag{103}$$

Writing $\mathcal{R}_{\ge 5}$ for the coefficientwise total-degree remainder in (93)*,*

$$\|\mathcal{R}_{\ge 5}[\mathscr{F}(E_1) - \mathscr{F}(E_2)]\|_{a_0;b} \le 6\left(\frac{C_H}{R} + C_{\text{quant}}\frac{gT}{b}\right)\|E_1 - E_2\|_{a_0;b}\,. \tag{104}$$

These estimates are uniform in old depth M and compatible volume. They assert a bound for the specified compatible canonical families, not for arbitrary localizations of the same summed functional.

Proof. By (98), the actual background difference is the family to which Theorem 4.1 applies. The coefficientwise jet condition gives

$$\|\mathcal{T}_{R,J}D\|_{a_0;b} \le \frac{C_H}{R}\|D\|_{a_0;b}\,. \tag{105}$$

The external $\Delta, Q_{\det}$ cancel coefficientwise. Linearity in (99) leaves

$$\mathscr{F}(E_1) - \mathscr{F}(E_2) = \mathcal{T}_{R,J}D + \mathcal{L}(Q_{E_2} + Q_D) - \mathcal{L}(Q_{E_2}).$$

If $D = 0$ the claim is immediate. Otherwise put

$$q_D = a_Q\|D\|_{a_0;b}\,, \qquad L_D = \frac{\eta_*}{32eq_D}. \tag{106}$$

Since $\|D\| \le 2\bar{\epsilon}$, (102) gives $q_D \le \eta_*/(64e)$ and $L_D \ge 2$. Interpolate the inherited exponent as $Q_{E_2} + \zeta Q_D$ on $|\zeta| < L_D$. Its total exponent with the fixed base satisfies

$$\|V_{\text{base}} + Q_{E_2} + \zeta Q_D\|_B \le \left(\frac{1}{64} + \frac{1}{128} + \frac{1}{32}\right)\frac{\eta_*}{e} = \frac{7\eta_*}{128e} < \frac{\eta_*}{8e}. \tag{107}$$

Equal-carrier exponentiation costs at most $e^x x$, where x is this bound. With $\eta_* \le 1$ its multiplier norm is at most $\eta_*/8$. The tail quarter budget then puts the connected-log norm at most $S_1\eta_*$ throughout the disk, by (72). Precision and base remain fixed, and the cluster theorem supplies one common nonvanishing logarithm branch.

The interpolated full input $E_2 + \zeta D$ need not stay inside the original $\bar{\epsilon}$ class on this larger disk. We have instead verified the actual Gaussian interaction budgets directly. Normal convergence gives a holomorphic anchored log family. Banach Cauchy, with the contour integral before the positive anchored sum, therefore yields

$$\begin{aligned} \|\mathcal{L}(Q_{E_1}) - \mathcal{L}(Q_{E_2})\|_{a_G;b} &\le \frac{S_1\eta_*}{L_D - 1} \le 64eS_1 q_D \\ &\le C_{\text{quant}} \frac{gT}{b} \|D\|_{a_0;b}\,, \qquad a_G = \frac{1}{2}. \end{aligned} \tag{108}$$

Both endpoints are evaluations of the same canonical analytic cluster family, so no independent logarithm branch or carrier choice is introduced. Since $a_G \ge a_0$, this proves (103). Finally (93) has norm at most six on the unchanged circular product ball and preserves the declared carriers. Applying it proves (104). □

5.3 A concrete one-half factor and its limitation

The parameter order can give a concrete projected factor. Fix the stronger common reserve and its C_A, κ_0, r_A, then choose a finite power $R = \ell^s$ satisfying all previous compression and amplitude conditions and

$$R \ge 24C_H. \tag{109}$$

Choose the common J and tail cutoff afterward, then one final b satisfying (97) and all classical and Gaussian neighborhoods. Finally choose $g, T(g)$ to meet (102) and

$$C_{\text{quant}} gT/b \le 1/24. \tag{110}$$

With a logarithmically growing cutoff the latter holds for all sufficiently small g at these fixed parameters. Equations (103)–(104) then give

$$\|\mathscr{F}(E_1) - \mathscr{F}(E_2)\|_{a_0;b} \le \frac{1}{12} \|D\|_{a_0;b}\,, \qquad \|\mathcal{R}_{\ge 5}[\mathscr{F}(E_1) - \mathscr{F}(E_2)]\|_{a_0;b} \le \frac{1}{2} \|D\|_{a_0;b}\,. \tag{111}$$

The same b is used at input and output. The canonical map above uses a common field-independent normalization. Separate output vacuum subtraction may double the unprojected estimate, but leaves the projected bound unchanged because $\mathcal{R}_{\ge 5}$ already kills constants.

The scalar sums compared here are differences of the actual common-domain quantum map. The proof first compares two admissible complete inputs; it does not apply a rooted/pre-gauge identity separately to a nongauge-invariant extraction. The representation-sensitive condition (96) and the compatible family construction are essential to the norm statement.

The full output difference can regenerate field jets of degrees zero through four through the fluctuation integral. Thus the quantum map does not preserve the affine input class with the prescribed low jets. Projecting the output is bounded but need not preserve gauge invariance, because gauge transformations are nonlinear in these potential coordinates. No gauge-invariant low-jet extraction or closed relevant/marginal flow has been constructed. This contractive response is therefore not yet a Banach fixed-point iteration on admissible physical activities. Large fields, mixed histories, global measure consistency, extraction and coupling flow, and continuum observables remain separate.

6 The next proof obligation

The finite step now returns the same analytic radius and the same positive connected-carrier exponent. Its amplitude estimate still has a coefficient larger than one multiplying the inherited activity norm. The background-dependent determinant and delta terms are also not small merely because the coupling is small. These statements locate the remaining small-field induction problem: extract the vacuum and allowed marginal terms, establish the coupling recursion, and prove a uniform bound on the remaining interactions through arbitrarily many steps.

The high-field-degree estimates provide one part of that recursion: coefficientwise fifth-order background remainders contract at a sufficiently large fixed macro ratio, and the actual quantum step has a contractive response to differences with matching low jets. Its Taylor subtraction is bounded without a radius loss. This does not give an invariant high-degree input class, because the fluctuation integral can regenerate lower-degree terms. The low-degree pieces still include kernels on arbitrarily many carrier geometries, and the separate pieces need not be gauge invariant. A complete extraction must combine the field-degree estimate with spatial locality and the nonlinear gauge identities, and include the generated low jets in the recurrence.

The total correction has the local gauge identity needed for that analysis. A scalar marginal gauge-action coefficient still requires the appropriate lattice symmetry classification; a Ward identity alone does not supply it. The large-field complements, mixed-history matching, observable estimates, continuum nontriviality and reconstruction remain separate required parts of the four-dimensional existence theorem.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 15–27 in this workspace. The homogeneous classical support families, strict overlapping Gaussian expansion, gauge identities, supported pre-gauge representative, auxiliary grouping and exact composite macrostep are the principal inputs.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Commun. Math. Phys. **116** (1988), 1–22. Original article copy.

[3] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.